

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

02nd November 2015

**Subject ST4 - Pensions and Other Employee
Benefits**

Time allowed: Three hours (14.45* – 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)**
- i)** Define sponsor covenant. (2)
 - ii)** Describe five methods of measuring the strength of sponsor covenant. (5)
 - iii)** Give one advantage and one disadvantage of each method identified in (ii) above. (5)
- [12]**

Q. 2) The Government of a country which currently does not provide any pension benefits is considering introducing a “National Pension Scheme” (NPS) for its residents.

- i)** List the areas where the Government may have a role in the provision of retirement benefits. (3)

The Government’s initial view is that on retirement, at age 60, subscribers must use the whole of their accumulated funds to purchase an annuity with an insurer. However this is expected to be unpopular with residents of the country because annuities are viewed as expensive and inflexible.

- ii)** Discuss the options that the Government should consider in order to address residents’ concerns. (5)

The Government decides to consult an Actuary. The Actuary proposes that residents with NPS scheme benefits should be able to take lump sum or retain up to 50% of their accumulated funds in their own personal investment account but will be required to purchase an annuity with the remaining amount. Residents would be able to withdraw money from their personal investment accounts to meet their on-going needs.

- iii)** Discuss the factors which may affect whether a resident chooses to take advantage of this increased flexibility. (8)

- iv)** Define the type of the scheme proposed by the Actuary. (1)

- v)** Write down the risks of the scheme to the Government and the residents who join the NPS. (4)

As an alternative, the Actuary also proposes a new scheme called “All Pension Plan” (APP). Under the APP, there is guaranteed minimum monthly pension ranging between Rs. 1000 and Rs. 5000 per month for the subscribers depending on the amount contributed by the subscribers prior to their retirement age.

- vi)** Define the type of the scheme proposed as an alternative. (1)

- vii)** Write down the risks of this alternative of scheme to the Government and the residents. (4)

Under APP, the benefits will be as follows:

- At Retirement, a fixed pension per month to the subscriber until his death.
- In case of death of the subscriber, a fixed pension would be available to the spouse for the rest of his/her life

As a general risk reduction exercise, the Government is thinking of buying insurance contracts to reduce the risk associated with this plan

viii) List the advantage and disadvantage of the insurance contracts that may be considered. (6)

ix) Outline the key features of the insurance contracts that may be considered. (6)
[38]

Q. 3) You are the actuary to the final salary pension scheme sponsored by a Company and are about to carry out a valuation of the Scheme. In 2012, the date of the last valuation of the Scheme, the value of the liabilities was significantly higher than the value of the assets. At the 2012 valuation, over half of the liabilities related to the eight senior employees of the Company.

In the course of a conversation with the finance director of the Company, she expressed the opinion that the funding position of the scheme is likely to have improved since 2012. The eight senior employees have suffered a cut of 50% in their basic salaries over the period and two years ago there was a redundancy exercise in one part of the business.

The active membership is back to its previous levels following a recruitment exercise in a different area of the business.

i) Describe how you would check the data to satisfy yourself that the details for members not yet retired are reliable. (5)

ii) Explain why past experience may be of limited relevance when setting the valuation assumptions for the benefits for members who are not senior employees. (8)

iii) Describe the investigations you would carry out before choosing the valuation assumptions for the senior employees' benefits. You are not required to consider how the discount rate might be determined.
Suggest, with reasons, suitable demographic assumptions for valuing the senior employees' benefits. (7)
[20]

Q. 4) A benefit scheme for the employees of a Company provides death in service cover of:

- lump sum of 4 times of basic salary; plus
- Pension for spouse of 25% of basic salary

i) List the principal assumptions required to estimate the cost of the benefits. (3)

ii) Outline the factors that need to be considered to set the assumptions. (12)

iii) Three life companies have quoted for insuring the death benefits. The three quotations vary considerably. Explain with reasons why the quotations could be so different. (5)
[20]

- Q. 5)** You are actuary to the defined benefit pension scheme sponsored by a company and recently carried out an actuarial valuation which has disclosed a large and unexpected surplus on the ongoing funding basis. Legislation in the relevant country requires that any valuation surplus needs to be eliminated over the next three years.

The trustees have asked you to set out the options available to eliminate the surplus.

Discuss the factors you would take into account and give examples of the different ways that the surplus could be used.

[10]
